

ORAL MUCOSA TUTORIAL

Dr. Misbah Ashraf Moten



TASK 1: PASTE THE FOLLOWING FIGURES FROM BERKOVITZ

**(NOTE: PLEASE WRITE CORRECT FIG. # FROM THE BOOK ALONG WITH
DESCRIPTION. MAKE COLORED DIAGRAMS)**

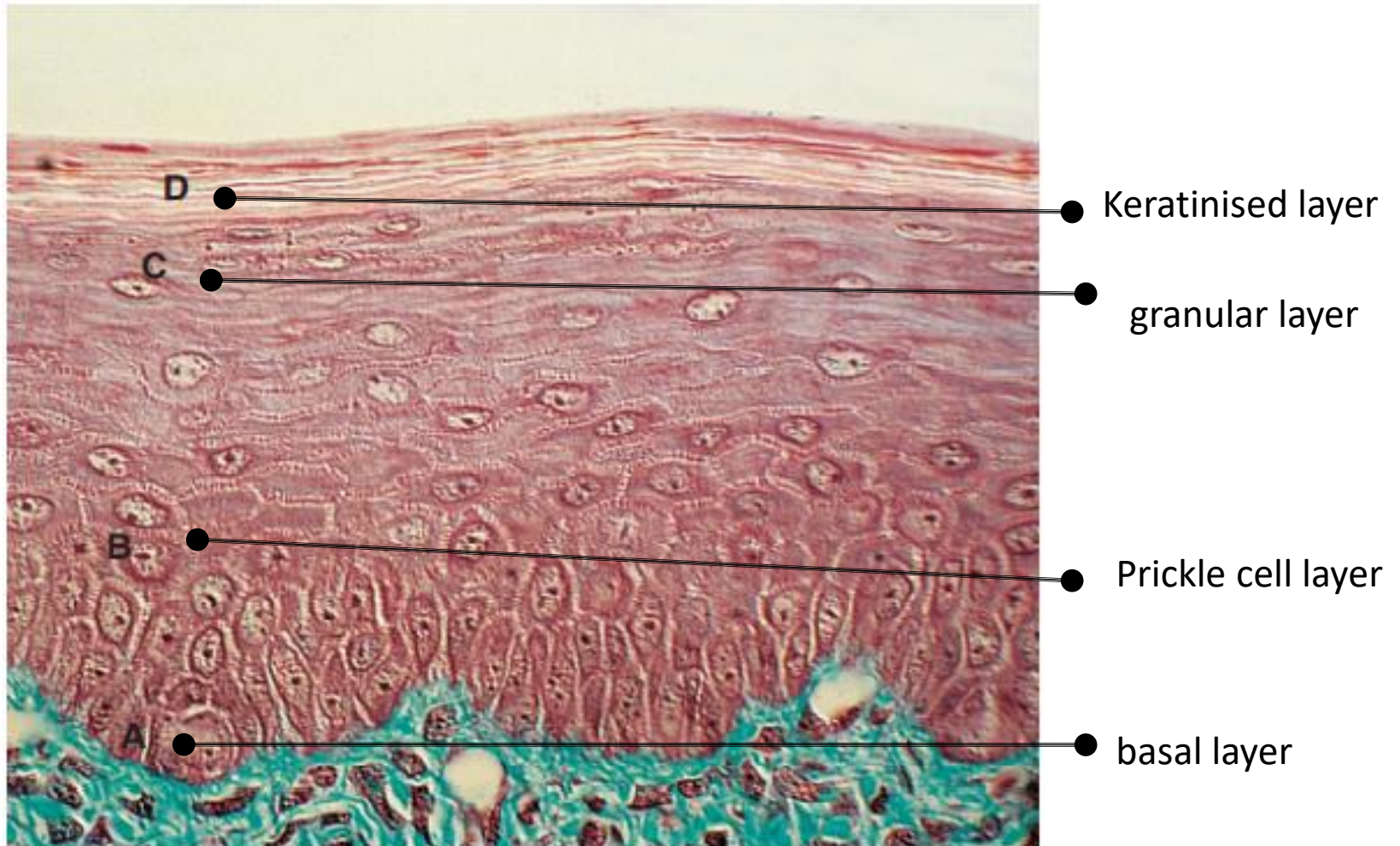


Fig. 14.2 Section showing layers of keratinised oral epithelium. A = basal layer; B = prickle cell layer; C = granular layer; D = keratinised layer ($\times 180$).

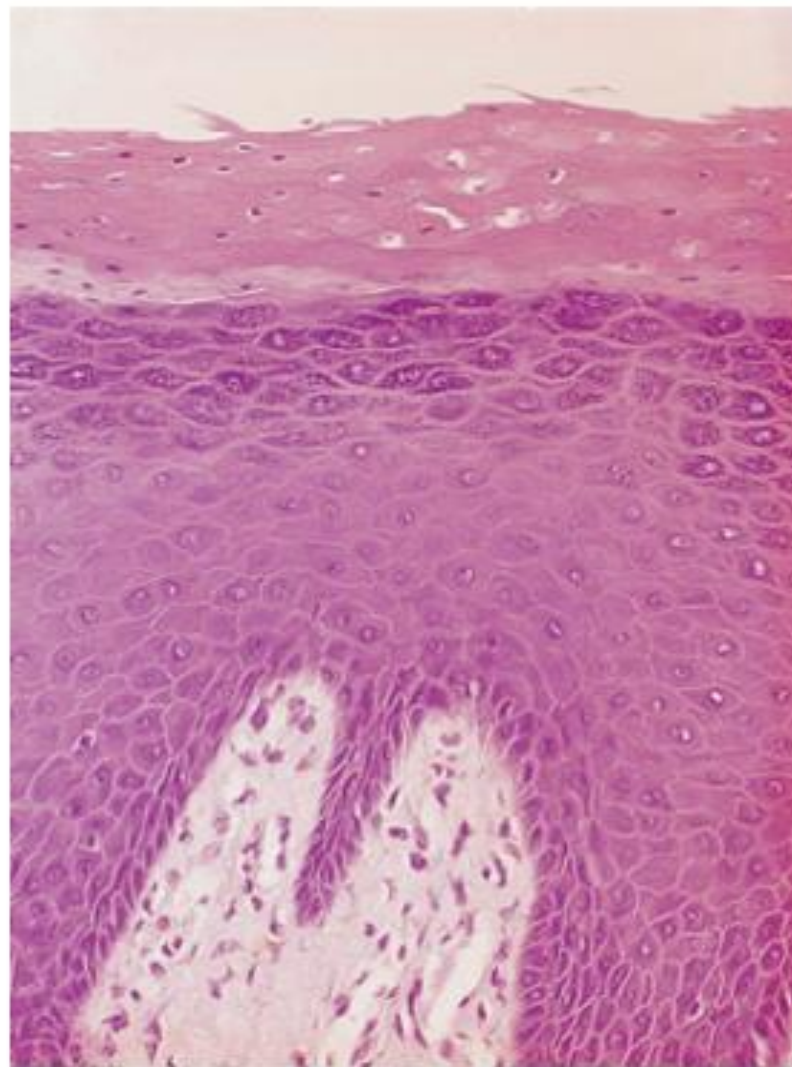


Fig. 14.12 Section of parakeratinised oral epithelium from the gingiva. The superficial layer stains more heavily for keratin but nuclei are retained (H & E; $\times 100$). Courtesy Dr A W Barrett.

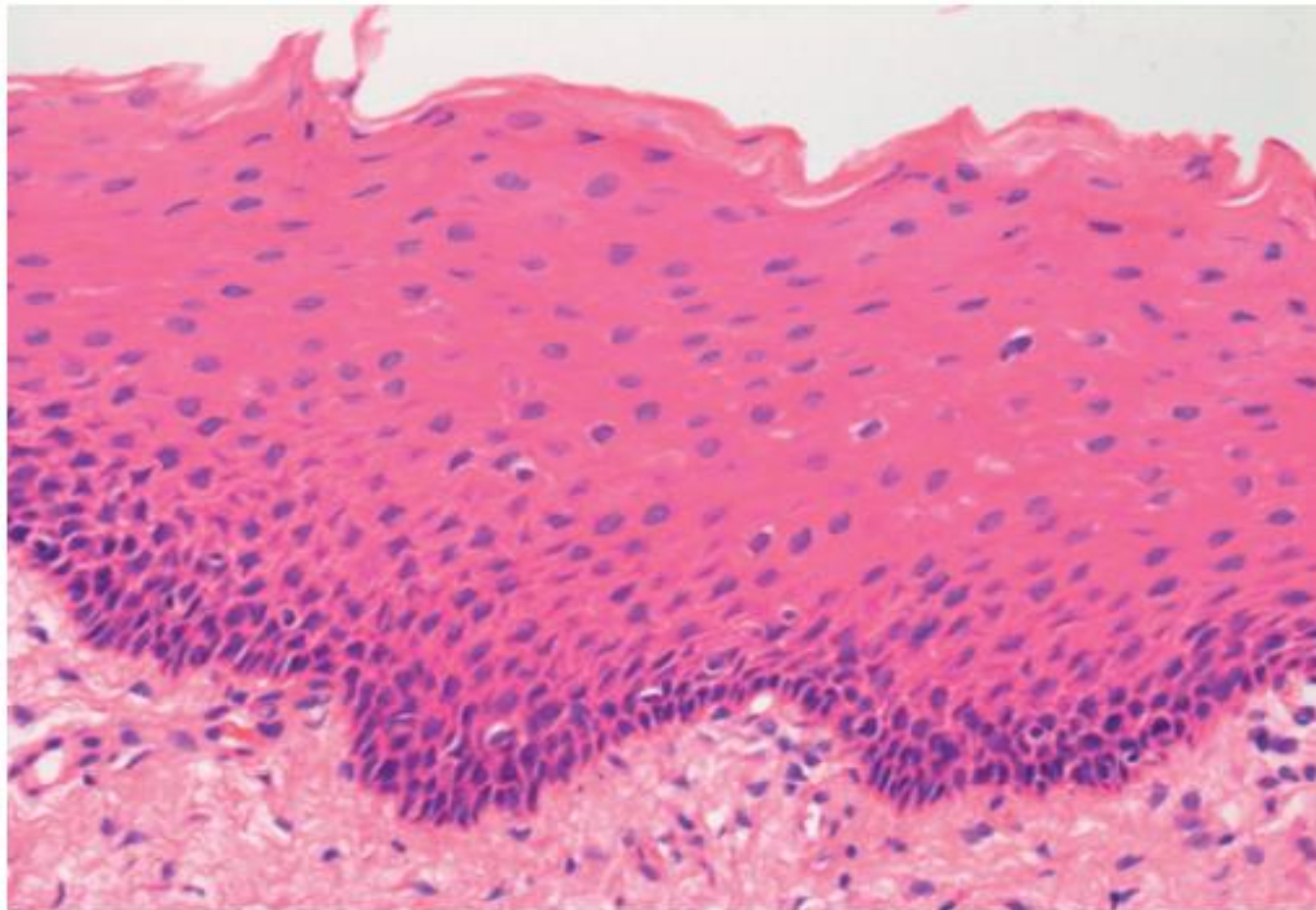


Fig. 14.13 Section of lining oral epithelium from the floor of the mouth. There is an absence of keratin in the superficial layer (H & E, $\times 150$). Courtesy Prof. P R Morgan.

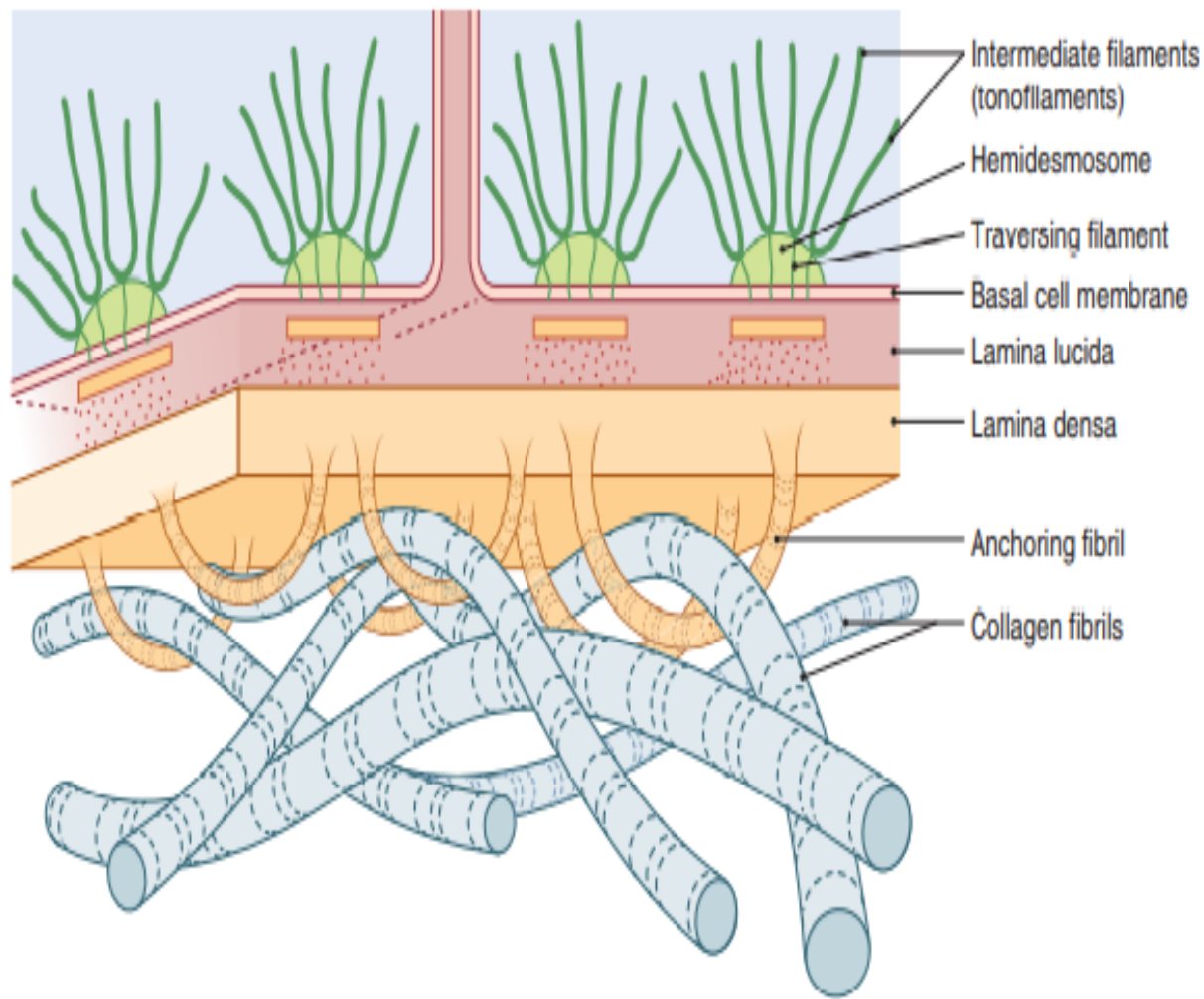


Fig. 14.29 Electron micrograph showing appearance of basal lamina ($\times 34,000$). Courtesy Prof. C A Squier.

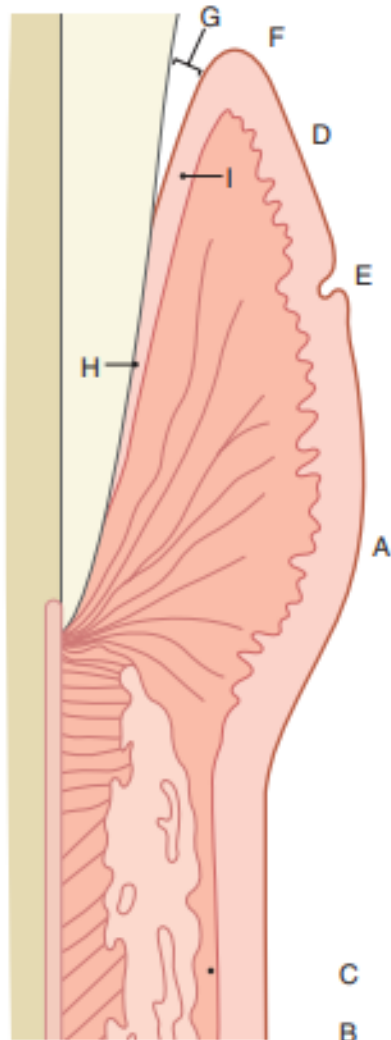


Fig. 14.36 Diagrammatic representation of gingiva. A = attached gingiva; B = alveolar mucosa; C = submucosa associated with alveolar mucosa; D = free gingiva; E = free gingival groove; F = gingival margin; G = gingival sulcus; H = junctional epithelium; I = crevicular epithelium.

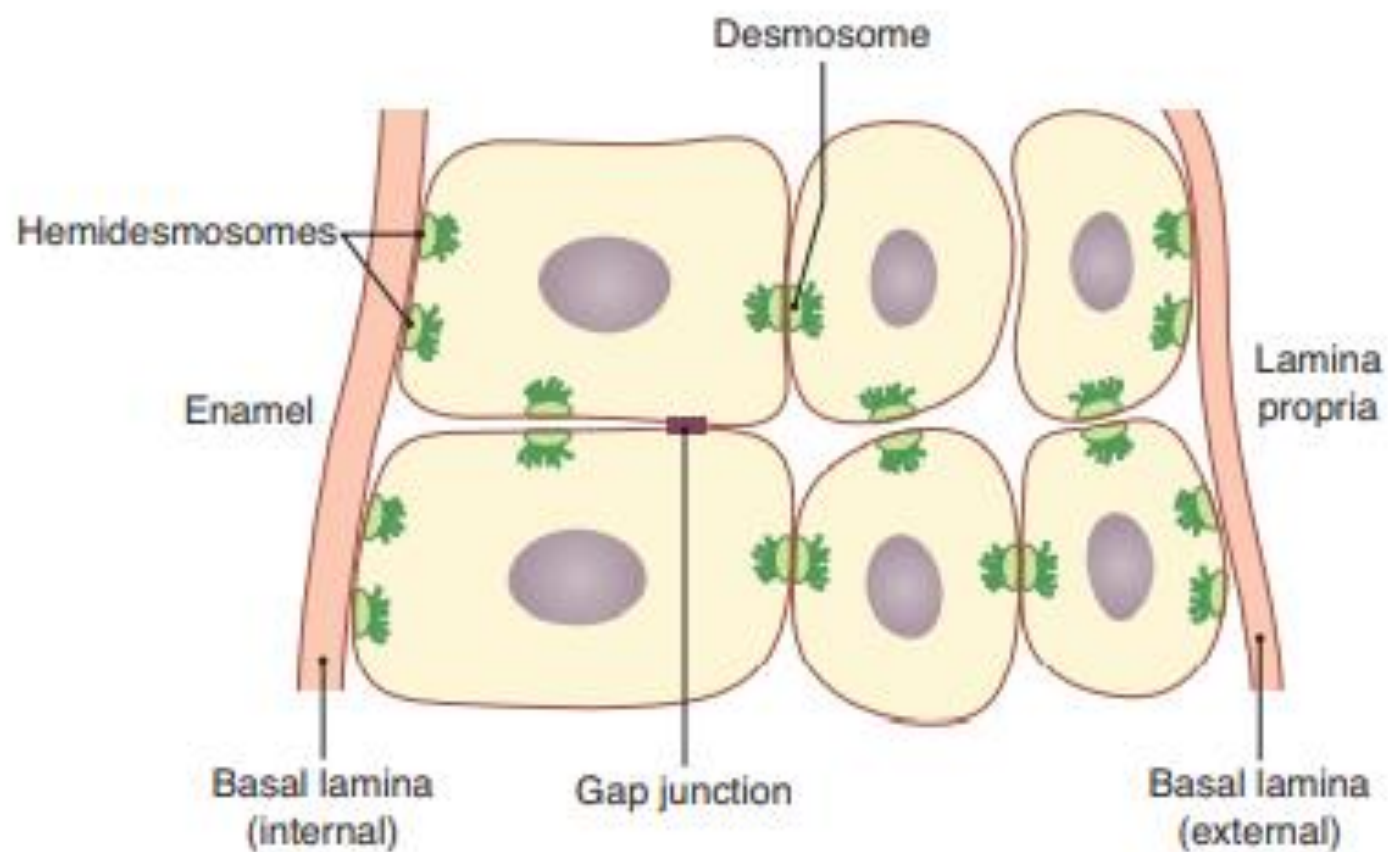


Fig. 14.46 Diagrammatic representation of junctional epithelium.

**TASK 2: COMPLETE QUESTIONS 1-5 IN
YOUR LOG BOOKS**

Q1. Differentiate between keratinized and non-keratinized Epithelium?

S. No.	Keratinized Epithelium	Non-Keratinized Epithelium
1	Has 4 layers: Basale, Spinosum, Granulosum, Corneum	Missing granular & keratinized layers; ends with superficial layer
2	Slower turnover (especially in masticatory mucosa)	Faster turnover rate
3	Keratohyalin granules present	Keratohyalin granules absent
4	Surface cells contain densely packed tonofilaments bound by filaggrin	Tonofilaments are loosely arranged and less developed
5	Surface cells may lose nuclei or retain small, shrunken ones in parakeratinized areas. Fewer organelles present.	Surface cells retain prominent nuclei and contain more organelles than keratinized type.
6	Well-defined stratified layers above the prickle cell layer	Upper layers are not clearly demarcated
7	Contains large intracellular membrane-coating granules ($\sim 0.25 \mu\text{m}$), rich in lipids and phospholipids; they form lamellar structures that create a strong permeability barrier	Contains smaller membrane-coating granules with less defined lipid layers, leading to increased permeability
8	Contains filaggrin, loricrin, and involucrin (important for keratinization)	Lacks filaggrin and loricrin, but has involucrin only

Q2. Enumerate the main functions and locations of different non-keratinocytes present in oral mucosa?

Ans. As many as 10% of the cells in the oral epithelium are non-keratinocytes & include

Cell names	Location	Function	Origin
Melanocytes	pigment-producing cells located in the basal layer	produce pigment melanin, using enzyme tyrosinase	Neural crest cells
Langerhans cells	dendritic cells situated in the layers above the basal layer	• Skin contact hypersensitivity reactions, • in antitumour immunity and in graft rejection; • propagators of human immunodeficiency virus (HIV)-1 transmission to T cells.	Bone marrow (monocyte lineage)
Merkel cells	basal layer often closely apposed to nerve fibres.	Act as a receptor.	Neural Crest
Inflammatory cells: • Lymphocytes (most common), • polymorphonuclear leukocytes & • plasma cells	Migrated through epithelium		

Q3. What is Junctional Epithelium? What are its features?

Ans. The junctional epithelium is an epithelial collar that surrounds the tooth and extends from the region of the CEJ to the bottom of the gingival crevice.

FUNCTION:

It serves to strengthen the seal of the dentognigival junction for underlying connective tissue of the periodontium from the oral environment and it may extend upto 2 mm.

Structure	- Two zones: single cell layer of cuboidal cells (stratum germinativum) over several flattened cells (stratum spinosum). - No stratum granulosum or corneum.
Cell Turnover	- Highest rate in oral mucosa (5-6 days). - Cells exfoliated coronally into the gingival crevice.
Origin	- Derived from rapidly replaced reduced enamel epithelium (probably from stratum intermedium component).
Basal Lamina	- Unique with two basal laminas: - Internal basal lamina (contacts tooth).- External basal lamina.
Cell Junction & Permeability	- Fewer desmosomes, larger intercellular spaces.- Increases permeability for GCF and defense cells.
Lamina Propria	- Rich blood supply in anastomosing network, with a rich plexus of nerves near the basal layer, many endings penetrating into the epithelium.

Q4. What are the different types of tongue papillae?

Type	Structure	Function
Filiform	Thin projections with central core of lamina propria & secondary branches.	Highly abrasive during mastication & provide friction; no taste buds.
Fungiform	Mushroom-shaped, scattered among filiform; 150–400 μm , thin epithelium.	Contain taste buds; involved in taste.
Foliate	1 or 2 longitudinal clefts at the side of the posterior part of the tongue	Taste buds in non-keratinized epithelium.
Circumvallate	Large papillae in a trench at the back of tongue; covered by non-keratinized epithelium.	Taste buds in internal trench walls; some Von Ebner's glands cleanse into these trenches.

Q5. Discuss the structure of a taste bud.

Ans. The taste buds, the special chemoreceptive organs responsible for taste, particularly found:

- around the walls of the circumvallate papillae
- in small numbers on the upper surface of fungiform papillae
- lateral walls of foliate papillae
- mucosa of the soft palate and in the epiglottis.

Structure of a Taste Bud

Each taste bud has a **small pore** at the surface, where taste molecules enter.

The bud is separated from deeper tissues by a **basal lamina (thin base layer)**.

Inside the taste bud are **two main types of cells**:

- **Supporting cells**
- **Taste cells**

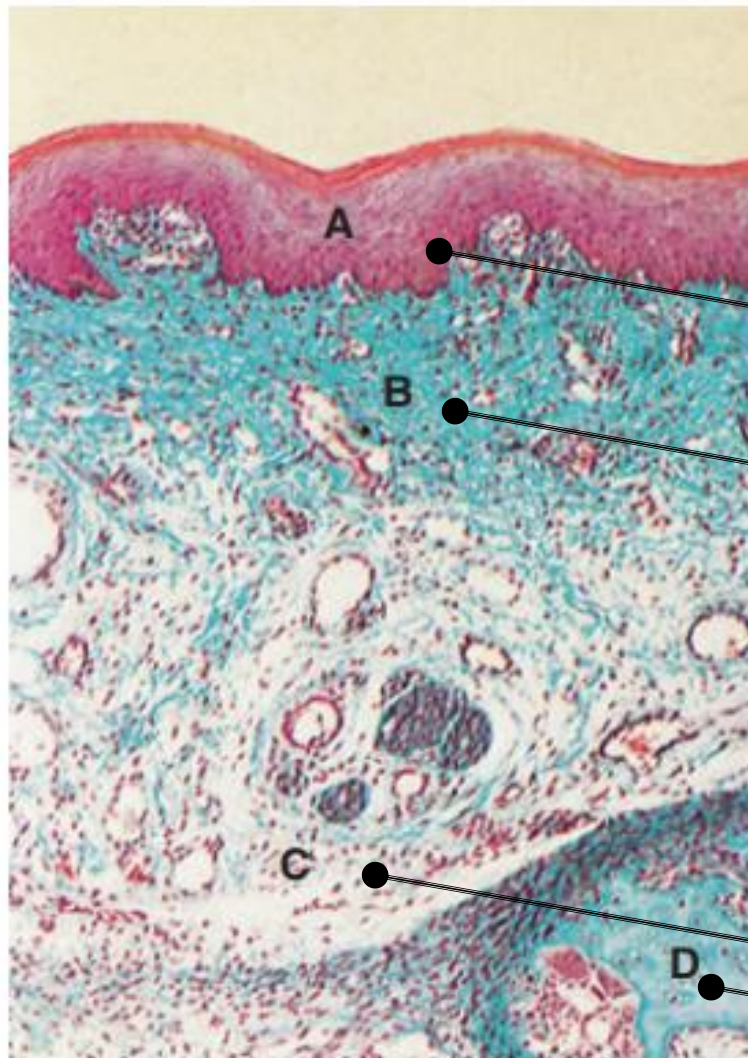
The morphology of a taste bud may vary according to species and site.

Four distinct cell types have been described:

Type of Cell	Function
Type IV cells	relatively undifferentiated cells which are distinguished by their basal position and the presence of intermediate filaments
Type I cells	have a dark appearance
Types II & III Cells	These cells are lighter.
Types I and III cells	may form synapses with intragemmal nerves.

IMPORTANT REFERENCE MATERIAL

Fig. 14.1 Section showing regions of oral mucosa. A = stratified squamous epithelium; B = lamina propria; C = submucosa; D = bone (Masson's trichrome; $\times 35$).

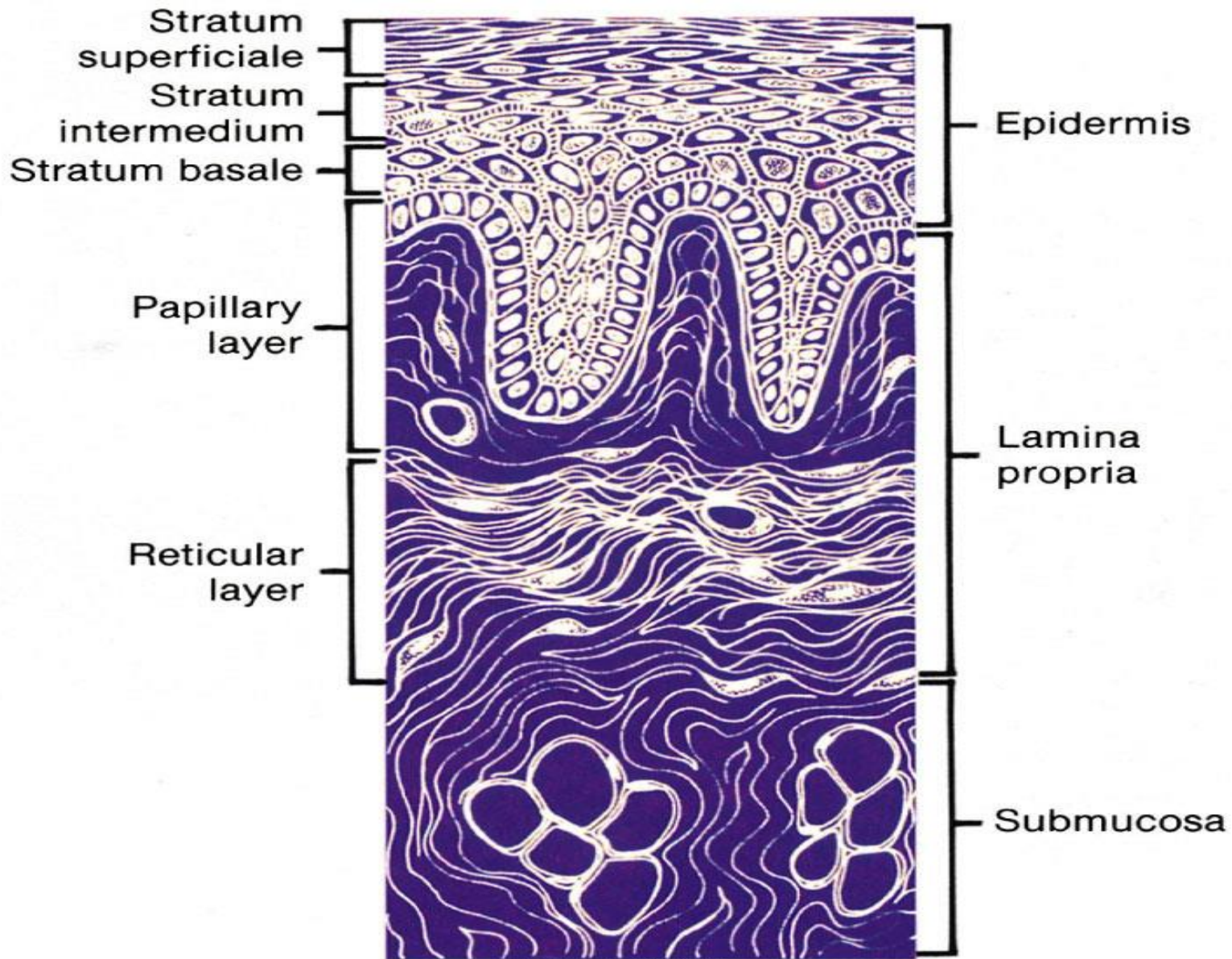


Stratified squamous
epithelium

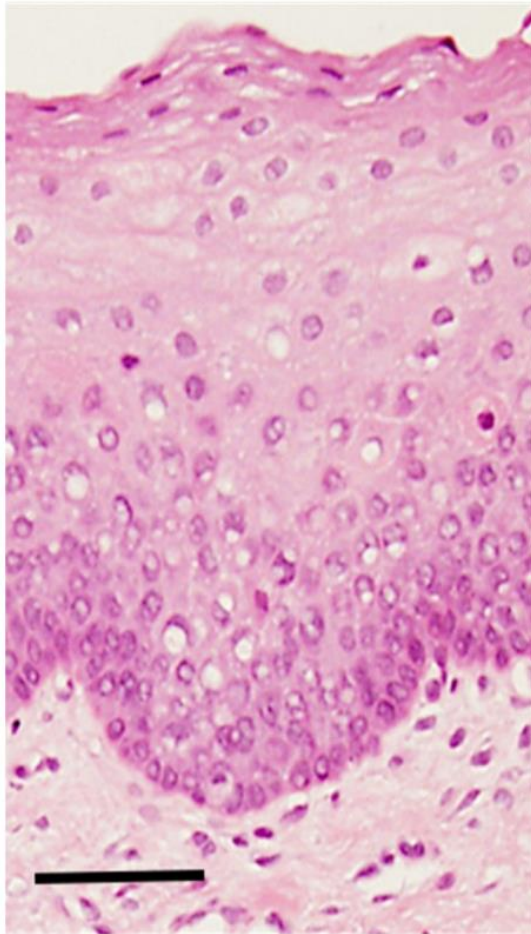
Lamina propria

Submucosa

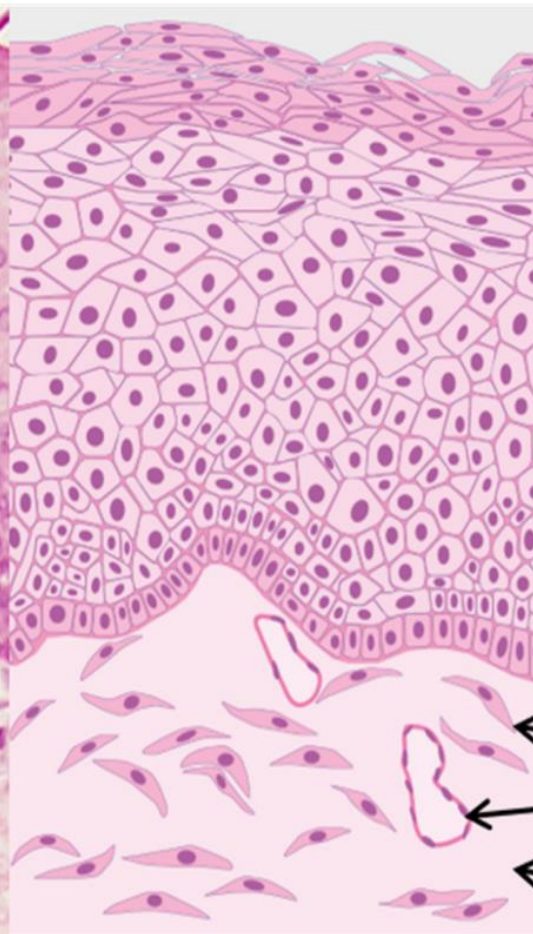
Bone



Histology



Schematic



Mucus

Superficial, nucleated
non-keratinized
differentiated epithelium

Stratum spinosum

Stratum basale

Oral fibroblast

Blood vessel

Connective tissue

Stratified squamous
epithelium

Lamina propria

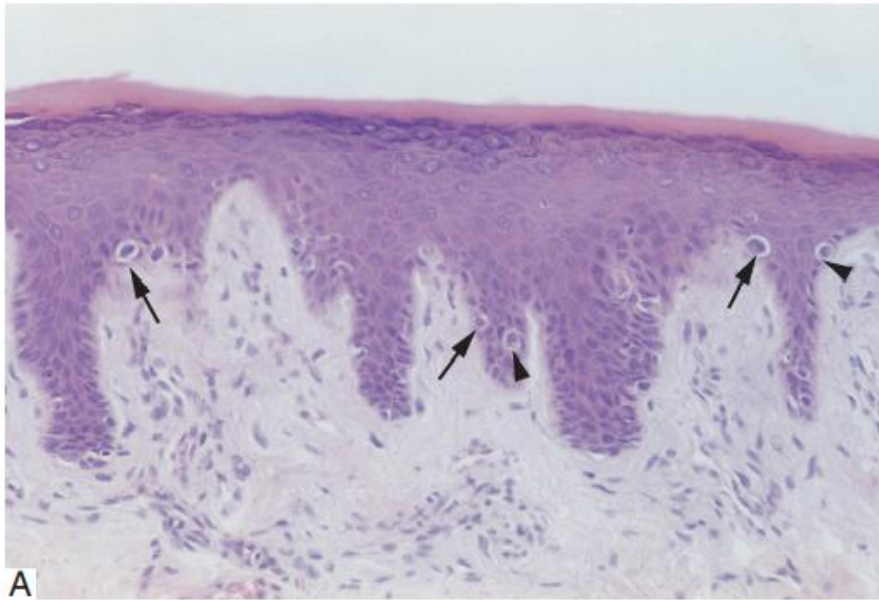
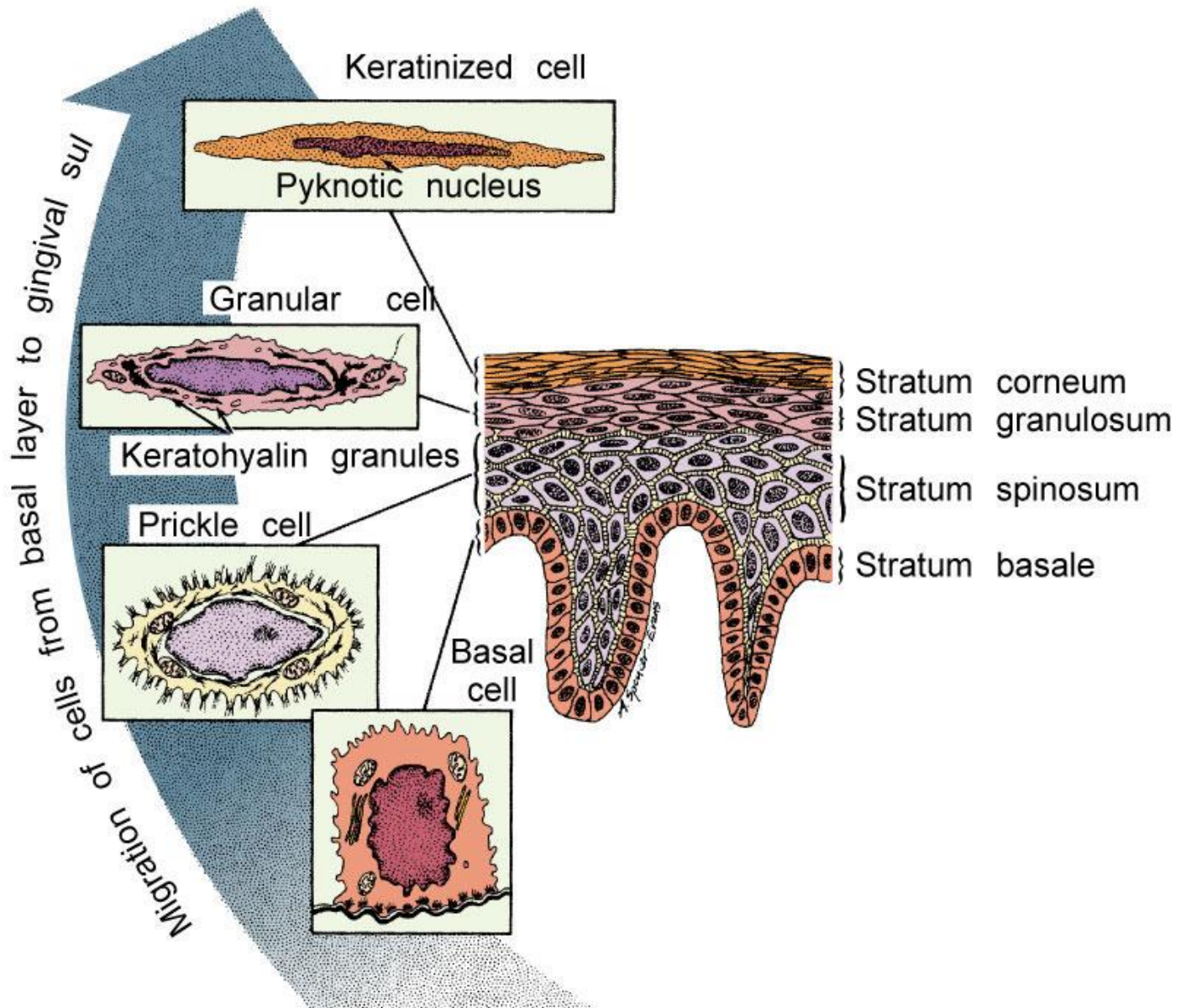


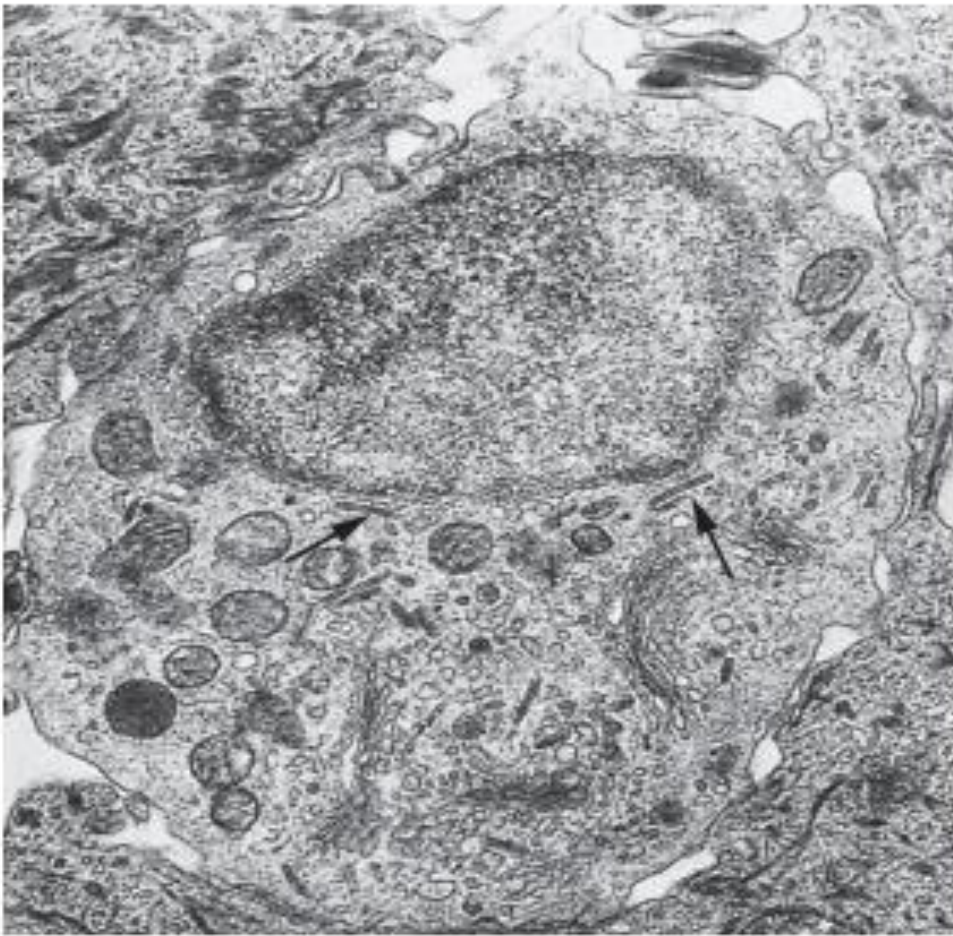
Fig. 14.19 (A) Section of the vermillion of lip showing clear cells (arrows) in the basal layer, representing mainly melanocytes ($\times 80$). (B) Clear cells in the upper layers of the epithelium (arrowed), mainly representing Langerhans cells (H & E; $\times 350$) Courtesy Prof. M M Smith.





D/f b/w Basement membrane and Basal Lamina

S. No.	BASEMENT MEMBRANE	BASAL LAMINA
1.	A complex arrangement links the surface epithelium to the underlying lamina propria in the oral mucosa. In the light microscope, a layer 1–2 μm thick is seen on the lamina propria side of the junction. This is termed the basement membrane.	Under the electron microscope, the layer appears much thinner and is then termed the basal lamina
2.	The cell membrane possesses specialized attachment plaques, the hemidesmosomes, along its length, where there is an increased density in structure. Cytoplasmic keratin filaments insert into the hemidesmosome.	The basal lamina is seen to consist of two layers): ■ an electron-lucent lamina lucida (20–40 nm thick) that lies immediately under the epithelium. ■ a thicker (20–120 nm) lamina densa.



Birbeck Granules:
Ultrastructurally, the Langerhans cell contains characteristic, trilaminar, rod-shaped granules called Birbeck granules. These may be up to 50 nm long and 4 nm wide with a vesicular swelling at one end, resembling a tennis racquet.

Fig. 14.24 Electron micrograph of Langerhans cell showing the characteristic Birbeck granules (arrows) ($\times 15,000$). Courtesy Dr N G El-Labban.

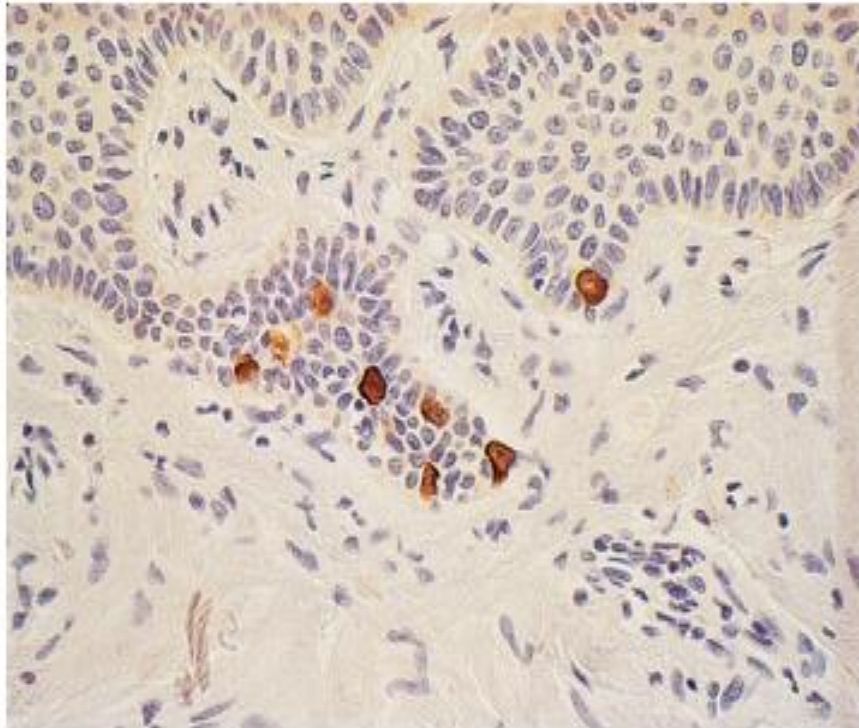


Fig. 14.25 Section showing Merkel cells in basal layer of epithelium staining positively (brown) for cytokeratin18 (streptavidin-biotinylated immunoperoxidase stain; $\times 200$). Courtesy Dr W A Barrett.

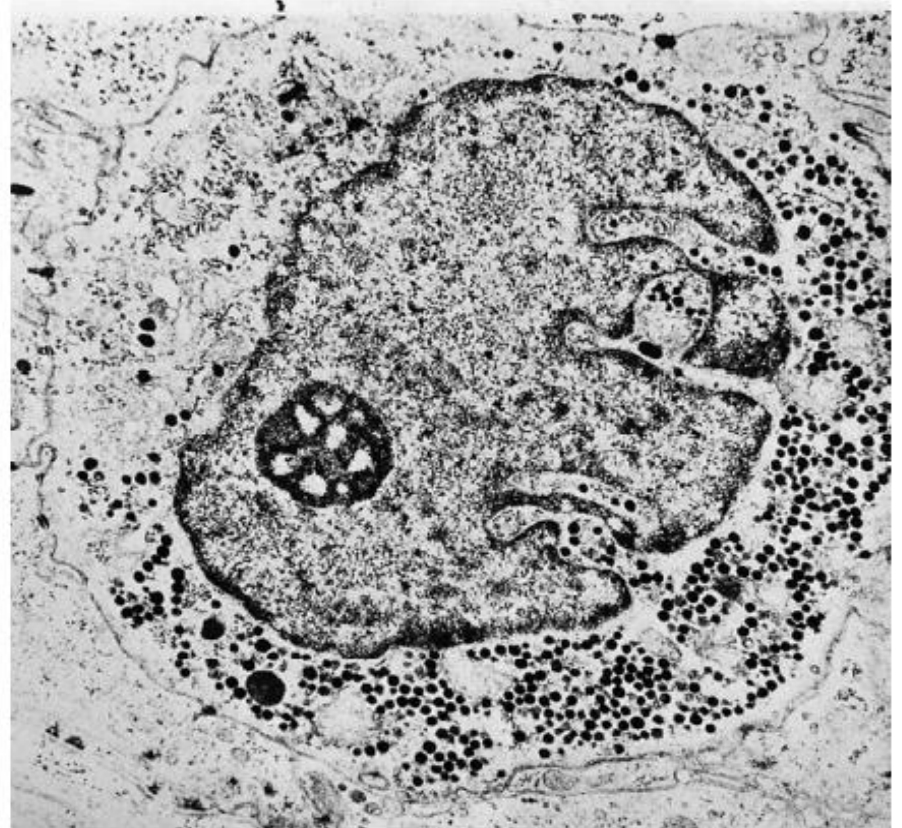


Fig. 14.26 Electron micrograph of Merkel cell. Note the indented nucleus and nuclear rodlet. The cytoplasm contains numerous characteristic dense granules 80–180 nm in diameter ($\times 16,000$). Courtesy Dr S-Y Chen.

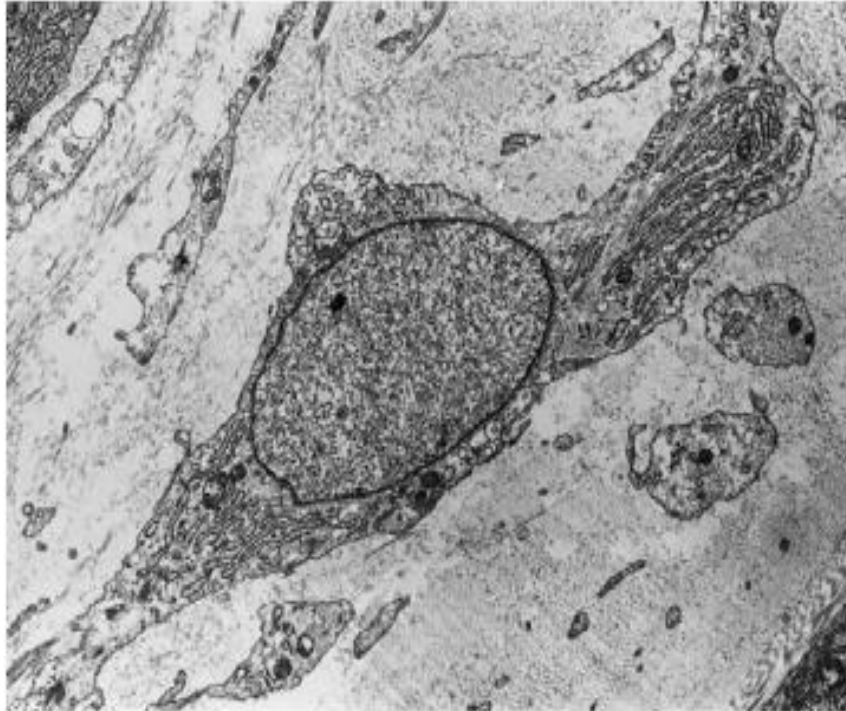


Fig. 14.27 Electron micrograph of fibroblast from lamina propria. The cell contains a full complement of all the intracellular organelles associated with collagen synthesis. Note the conspicuous rough endoplasmic reticulum ($\times 11,000$). Courtesy Prof. C A Squier.

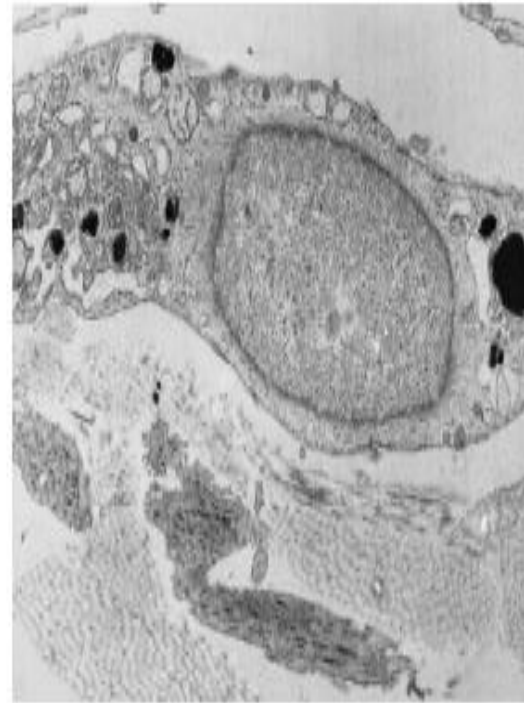


Fig. 14.28 Electron micrograph of macrophage in lamina propria. It contains many lysosomes but little endoplasmic reticulum ($\times 10,000$). Courtesy Prof. C A Squier.

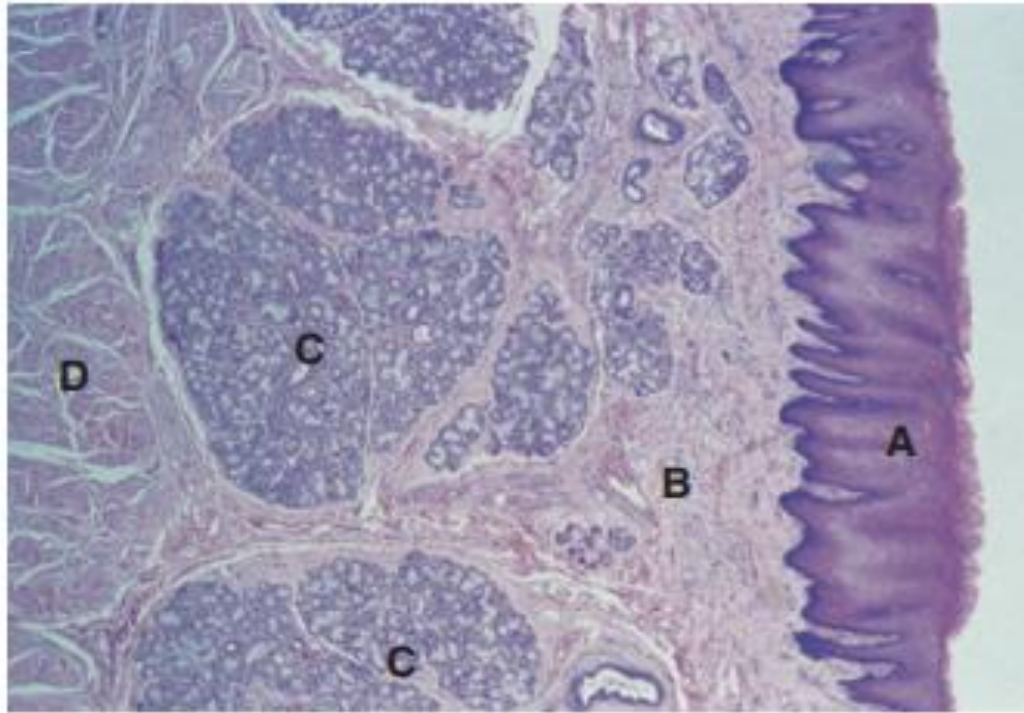


Fig. 14.33 Section of labial mucosa. A = nonkeratinised oral epithelium; B = lamina propria; C = minor salivary gland in submucosa; D = fibres of orbicularis oris (H & E; $\times 15$). Courtesy S Kariyawasam.

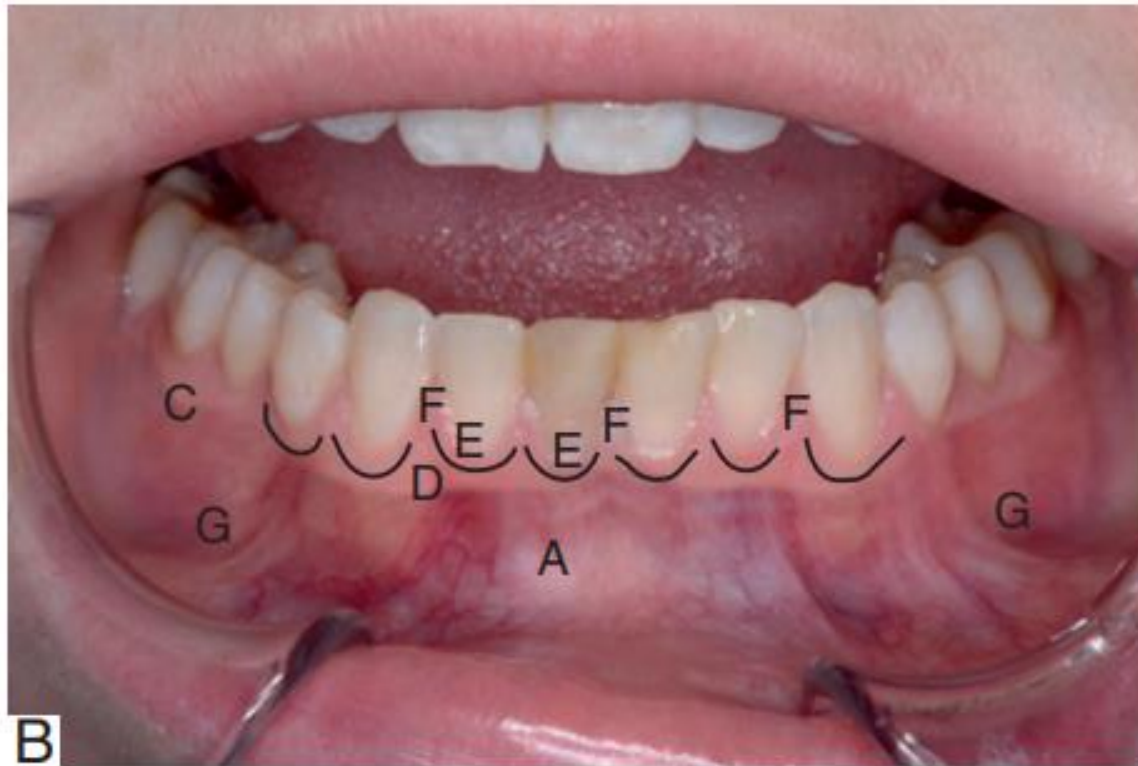


Fig. 14.37 *In vivo* appearance of maxillary gingiva (A) and mandibular gingiva (B). A = alveolar mucosa; B = free gingival groove indicated by curved black line; C = mucogingival junction; D = attached gingiva; E = free gingiva; F = interdental papilla.

Table 14.2 Principal features and regional variations of the oral mucosa

Region	Epithelium		Lamina propria		Submucosa		Type of mucosa
	Thickness	Keratinisation	Papillae	Fibre types	Density	Attachment	
Labial and buccal mucosa	Thick	Nonkeratinised	Short and irregular	Collagen and some elastin fibres	Dense	Firmly to underlying muscle	Lining
Vermilion (red) zone of lip	Thin	Keratinised	Long and narrow	Collagen and some elastin fibres	Dense	Firmly to underlying muscle	Specialised
Alveolar mucosa	Thin	Nonkeratinised	Short or absent	Many elastin fibres	Loose	Loose attachment to periosteum	Lining
Attached gingiva	Thick	Keratinised and parakeratinised	Long and narrow	Dense collagen firmly attached to underlying periosteum	No distinct submucosa		Masticatory
Floor of mouth	Thin	Nonkeratinised	Short and broad	Collagen and some elastin fibres	Loose	Loose attachment to underlying muscle	Lining
Ventral surface of tongue	Thin	Nonkeratinised	Short and numerous	Collagen and some elastin fibres	Not very distinct layer; attached to underlying muscle		Lining
Dorsum of tongue (anterior two-thirds)	Thick	Primarily keratinised	Long	Collagen and some elastin fibres	Not very distinct layer; attached to underlying muscle		Specialised gustatory
Dorsum of tongue (posterior one-third)	Variable	Generally nonkeratinised	Short or absent	Collagen and some elastin fibres	Not very distinct layer; attached to underlying muscle		Lining gustatory
Hard palate	Thick	Keratinised	Long	Dense collagen in submucosa laterally, but lamina propria firmly bound to periosteum without submucosa in midline			Masticatory
Soft palate	Thick	Nonkeratinised	Short	Many elastin fibres	Loose	Loose attachment to underlying tissues	Lining

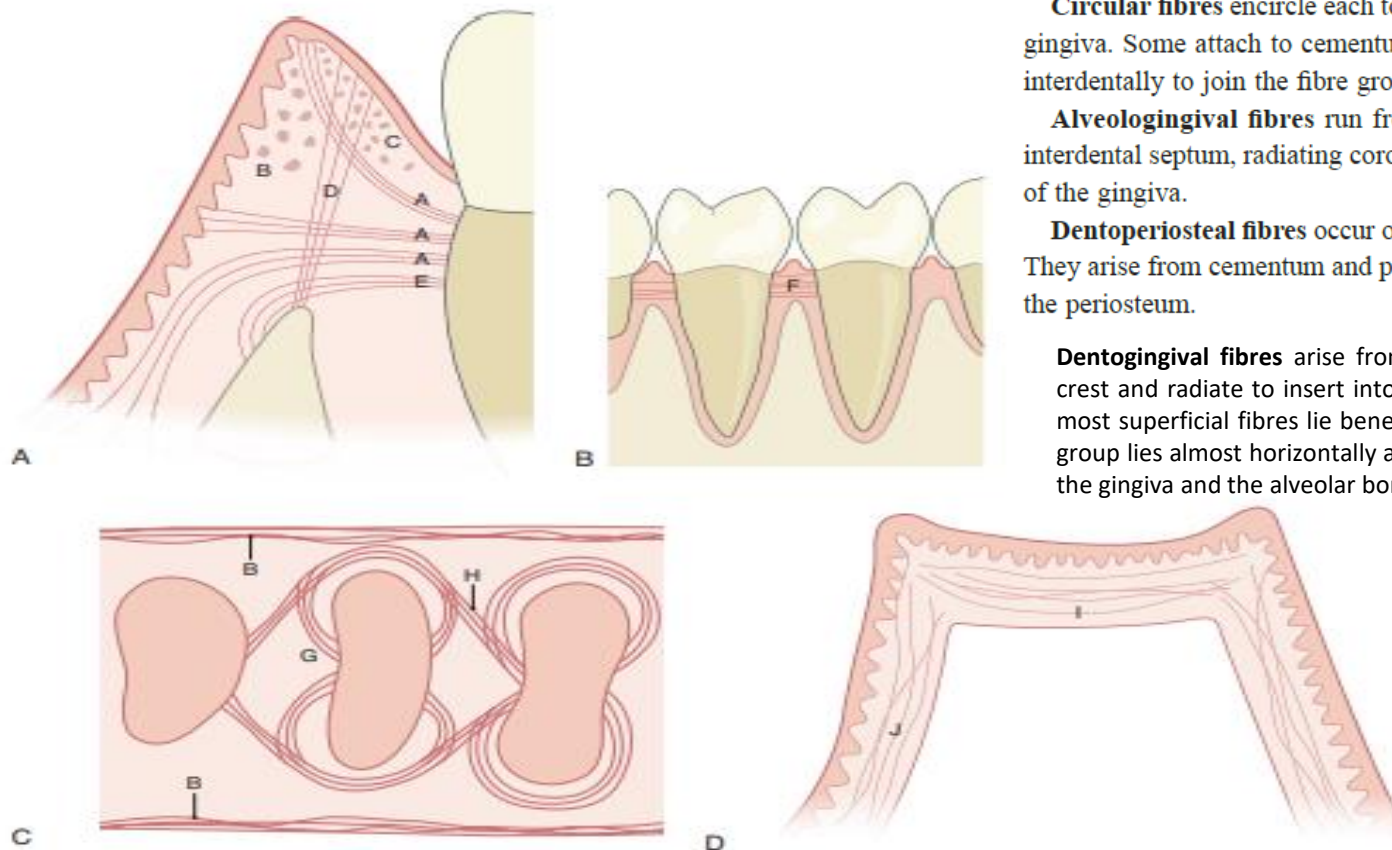


Fig. 14.55 Diagram showing arrangement of principal collagen fibre groups in the lamina propria of the gingiva. (A) buccolingual section; (B) mesiodistal section; (C) horizontal section; (D) buccolingual section along interdental col. A = dentogingival fibres; B = longitudinal fibres; C = circular fibres; D = alveologingival fibres; E = dentoperiosteal fibres; F = transseptal fibres; G = semicircular fibres; H = transgingival fibres; I = interdental fibres; J = vertical fibres.

Longitudinal fibres extend for long distances within the free gingiva, some possibly for the whole length of the arch.

Circular fibres encircle each tooth within the marginal and interdental gingiva. Some attach to cementum, some to alveolar bone. Some cross interdentrally to join the fibre group of the adjacent tooth.

Alveologingival fibres run from the crest of the alveolar bone and interdental septum, radiating coronally into the overlying lamina propria of the gingiva.

Dentoperiosteal fibres occur only in labial/buccal and lingual gingiva. They arise from cementum and pass over the alveolar crest to insert into the periosteum.

Dentogingival fibres arise from the root surface above the alveolar crest and radiate to insert into the lamina propria of the gingiva. The most superficial fibres lie beneath the crevicular epithelium, a middle group lies almost horizontally and the deepest group courses between the gingiva and the alveolar bone

Transseptal fibres pass horizontally from the root of one tooth, above the alveolar crest, to be inserted into the root of the adjacent tooth (Figs 14.52, 14.57). Such fibres provide an anatomical basis for linking all the teeth in the dentition. They have been implicated in the mechanism of mesial drift (page 437).

Semicircular fibres emanate from cementum near the cementum–enamel junction, cross the free marginal gingiva and insert into a similar position on the opposite side of the tooth.

Transgingival fibres reinforce the circular and semicircular fibres. The fibres arise from the cervical cementum and extend into the marginal gingiva of the adjacent tooth, merging with the circular fibres.

Interdental fibres pass through the coronal portion of the interdental gingiva in the buccolingual direction, connecting buccal and lingual papillae.

Vertical fibres arise in alveolar mucosa or attached gingiva and pass coronally towards the marginal gingiva and interdental papilla.

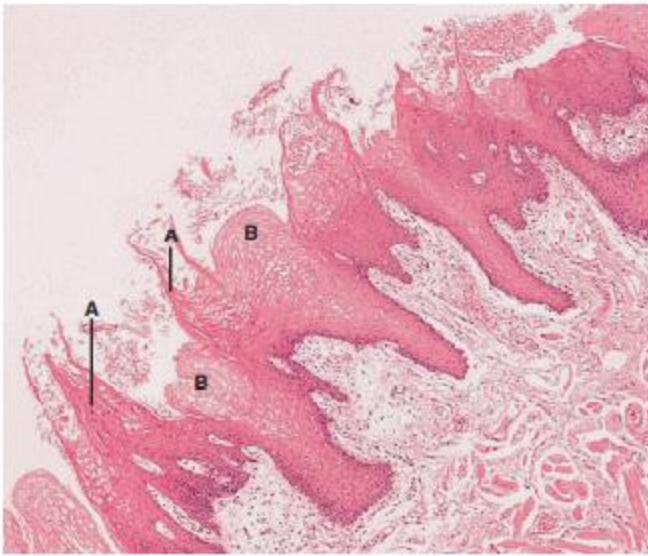


Fig. 14.66 Section showing dorsum of anterior two-thirds of tongue covered by keratinised filiform papillae (A) with nonkeratinised regions between (B) (H & E; $\times 35$). Courtesy Prof. P R Morgan.

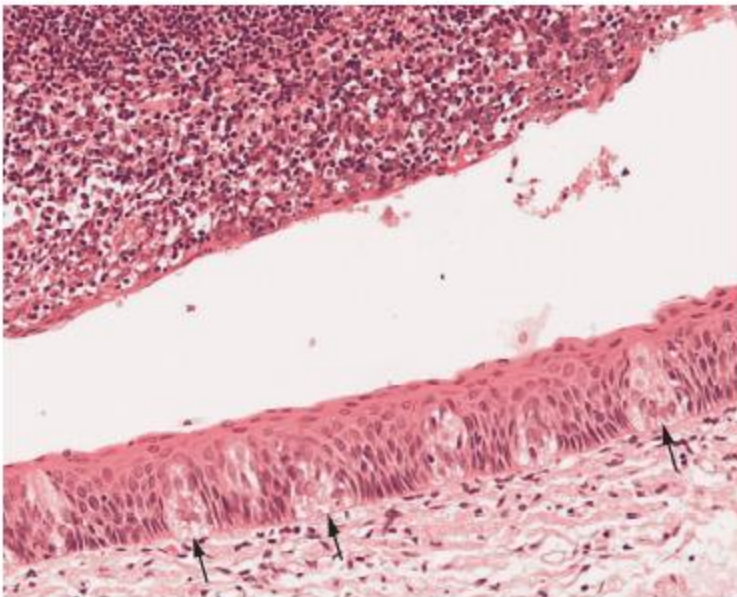
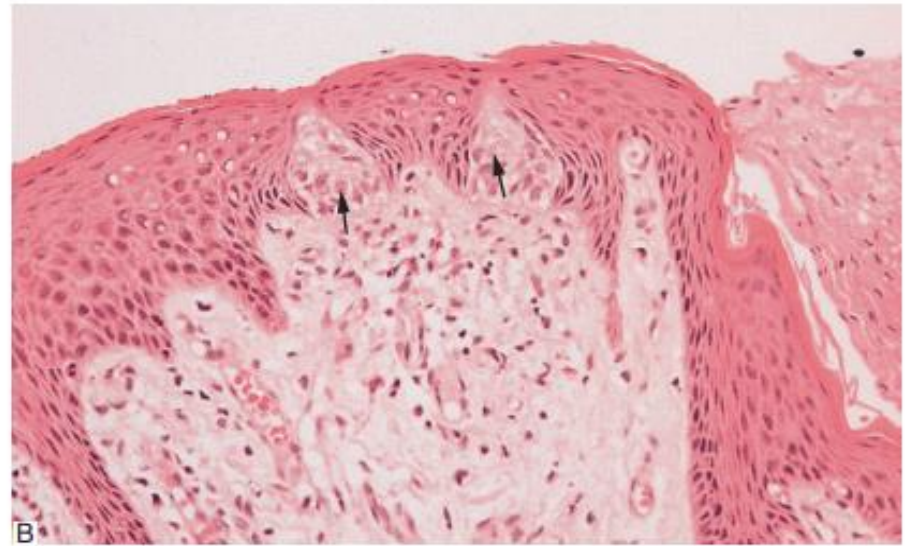


Fig. 14.70 Section of foliate papilla showing taste buds (arrowed). Note the adjacent lymphoid material characteristic of the posterior part of the tongue ($\times 200$).

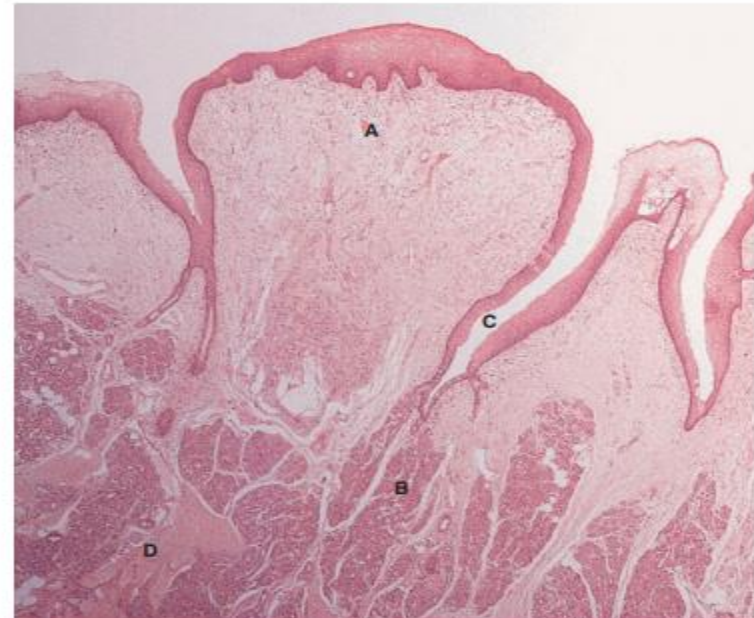
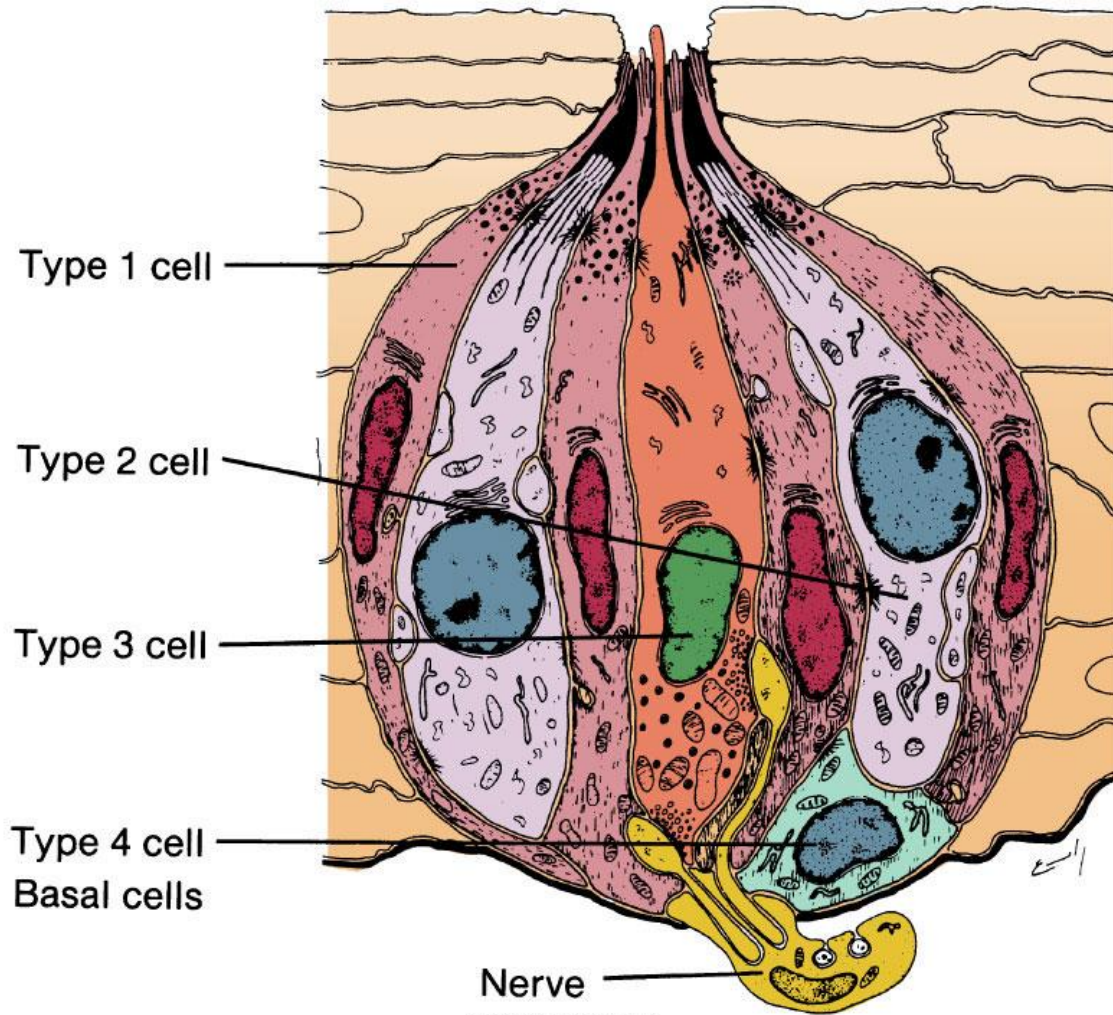
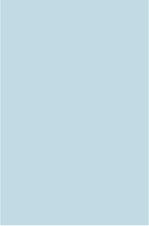


Fig. 14.72 Section of circumvallate papilla (A). Serous glands (B) of von Ebner empty via ducts into the base of the trench (C) surrounding the papilla, which is not raised above the surface of the tongue. D = muscle of tongue (H & E; $\times 35$). Courtesy Prof. P R Morgan.



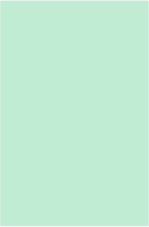
CYTOKERATINS & ITS CHARACTERISTICS



CK 6 & 16: rapidly turnover (hyper-proliferative) epithelia



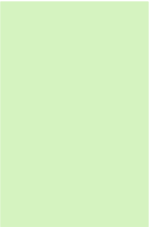
CK 5, 6 & 14: epithelium lining ventral surface of tongue



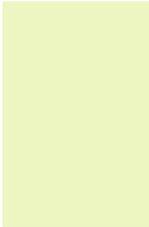
CK 19: basal keratinocytes in lining mucosa (inconsistent)



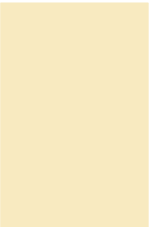
CK 4 & 13: suprabasal keratinocytes of lining mucosa



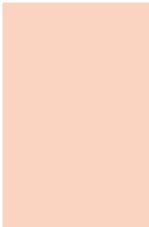
CK 7, 8, 18 & 19: epithelium covering soft palate



CK 5 & 14: restricted to basal & parabasal layers



CK 1, 2, 10 & 11 : suprabasal layers of masticatory mucosa



CK 8 & 18: simple epithelium



Fig. 14.78 View of back of tongue showing a lingual thyroid (arrow). Courtesy Prof. J D Langdon.



Fig. 14.85 Lichen planus in the buccal mucosa, producing a lacy network of white striae. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.



Fig. 14.86 Squamous cell carcinoma showing a speckled appearance, firmness on palpation (induration) and an ulcerated rolled margin. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.



Fig. 14.80 (A) Typical appearance of erythema migrans (geographic tongue) showing irregular depapillated red patches centred on the lateral sides of the tongue, with each patch having a narrow red and white rim. (B) Tongue of the same patient shown in Fig. 14.80A but taken on a later occasion and showing a change in the pattern of red patches. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.

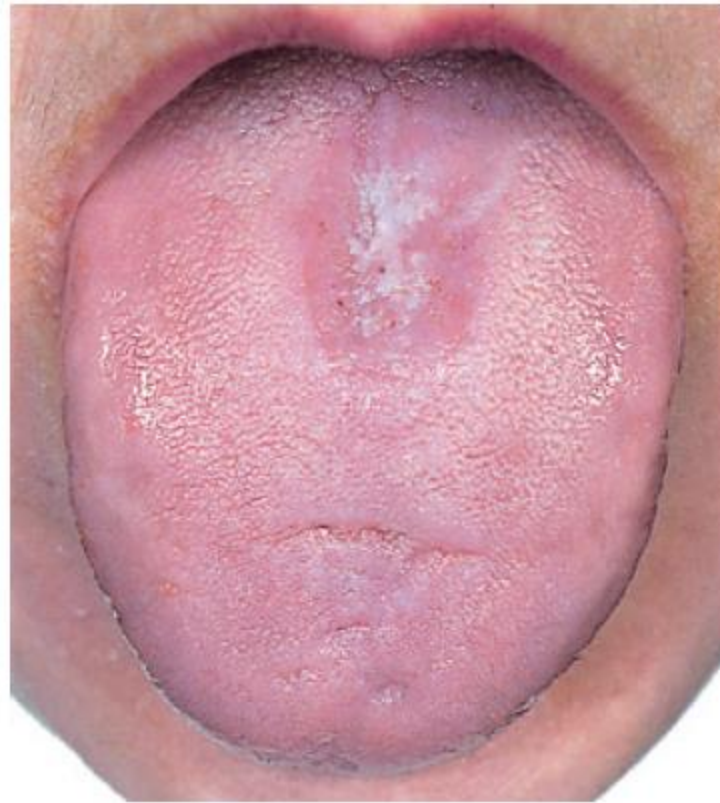


Fig. 14.82 The typical lozenge-shaped area of depapillation in the midline of the dorsum of the tongue representing median rhomboid glossitis. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.